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PROTOCOL

STrategies for developing REseArch Methods guidance (STREAM): Protocol

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Abstract

Background: Health researchers rely on clear, trustworthy methods guidance (MG) to support the planning, conduct, analysis, and interpretation of their studies (eg, guidance on study design, data collection, statistical methods, qualitative analysis). MG should not only be practically useful but also help ensure that the resulting evidence is valid and meaningful to patients, research participants, and other interest-holders. However, how to develop such methods guidance and which processes and approaches are useful is currently unclear. Important uncertainties include how to integrate methodological evidence and the views of health researchers, the intended users. This lack of clarity in the development process may undermine the validity, trustworthiness, and usefulness of MG, ultimately limiting its contribution to the production of high-quality, patient-important evidence.

Objectives: To describe STrategies for developing REseArch Methods guidance (STREAM).

Methods: We will work with a steering committee consisting of MG developers, health researchers, and an experienced patient partner who will help us establish a deliberative and meaningful patient contribution. To systematically develop STREAM, we will conduct a series of interlinked studies: (1) Review of existing standards: A scoping review to identify any existing standards or suggestions for developing MG. (2) Study of current practice: We will analyse a sample of recently published MG articles and catalogue the methods used for developing MG. (3) Interviews with MG users: We will conduct interviews with recent users of selected MG articles to elicit their views on strengths, weaknesses, and opportunities for improvement. (4) Development of STREAM: A panel of guidance developers and users from academia and industry will draft an initial version of STREAM and apply consensus methods. The panel will consider the findings from studies 1-3 and processes for developing other types of guidance, such as reporting guidelines and clinical practice guidelines. (5) Testing and refining STREAM: Guidance developers will apply the draft STREAM to ongoing MG development projects and provide feedback that will inform the final version.

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Results: STREAM will be a comprehensive, evidence-informed, user-centred, and practice-tested instrument for creating MG. It will provide practical recommendations for MG developers on involving relevant interest-holders, selecting and using methodological evidence, translating evidence into recommendations, and presenting guidance in a clear and implementable format.

Keywords: Methods guidance; Evidence-based research practice; Quality of research; Methods review; Meta-research; Interview study; Consensus study

Plain Language Summary

Health researchers need clear and trustworthy guidance to help them plan, conduct, analyse, and interpret their studies. The guidance should ensure that the research is valid and genuinely beneficial to patients. However, there is currently uncertainty about the ways for creating this guidance. As a result, experts often develop guidance in an unstructured way or are uncertain how to integrate existing evidence or the views of health researchers who are the intended users of the guidance. This may undermine trust in the guidance, making it less helpful than it could be. Our goal is to address this. We aim to provide a process, STREAM, for creating trustworthy and user-friendly guidance that leads to meaningful and valid health research. Here is our plan: (1) Review existing suggestions for creating guidance: We will analyze existing publications that make suggestions on how to create guidance. (2) Review existing guidance: We will analyse a sample of recent guidance to see how experts currently create guidance: for example, how they collected information, involved users, reached a consensus, and tested the guidance. (3) Engage with guidance users: We will interview health researchers who recently used guidance. We want to understand what makes guidance trustworthy and meaningful to them. (4) Develop STREAM: We will assemble a diverse group including experts and health researchers. Together, we will develop the new process, STREAM. (5) Test STREAM: We will collaborate with experts currently creating guidance and ask them to test the new process. We will collect feedback to refine and finalize STREAM. To ensure that our research makes sense, we will consider possibilities for engaging health researchers, patients, and members of the public throughout the projects. In summary, our aim is to provide practical advice for experts who create guidance for conducting health research. We will call the new process STREAM and it will be evidence-based, user-centred, and practice-tested. STREAM will make it easier to create guidance that is clearer for health researchers and will lead to evidence that is more actionable and relevant to patients.

1. Introduction

High-quality evidence is fundamental for optimal health care. Much of the evidence, however, is constrained by avoidable methodological limitations, leading to biased findings [1−3], waste of resources [4−7], and ethical concerns [8,9].

Many meta-studies have systematically assessed the methodological quality of clinical studies (eg, regarding the handling of missing data, subgroup analyses, or patient-reported outcomes). They are strikingly consistent in concluding that important methods in clinical studies are often substandard and that the implementation of available methods is slow and patchy [3,4,10−13]. For instance, a review of 142 randomized clinical trials found flaws in the areas of randomization, blinding, and missing data in 96% of the studies [10]. The authors also noted that using optimal methods would often have been possible and usually at no additional cost [10]. More recent studies on clinical trials [3], prognostic models, [14] observational studies [15], and clinical practice guidelines [16] came to very similar conclusions.

Avoidable methodological flaws pose serious ethical concerns. The resulting biases and uncertainties can lead to inappropriate therapy choices and may eventually harm patients [17,18]. Many researchers therefore argue that rigor and methodology need to be at the heart of health research [7]. As a result, the recently updated Declaration of Helsinki that defines the fundamental ethical principles for research involving humans now explicitly requires optimal methods: *"Medical research involving human participants must have a scientifically sound and rigorous design and execution that are likely to produce reliable, valid, and valuable knowledge and avoid research waste. The research must conform to generally accepted scientific principles, be based on a thorough knowledge of the scientific literature, other relevant sources of information, and adequate laboratory and, as appropriate, animal experimentation."* [19,20].

Written methods guidance (MG) plays a key role in articulating such scientific principles for health research [21]. Written MG exists in different formats including published peer-reviewed journal articles, for example, in the Journal of the American Medical Association [22,23] and the British Medical Journal [24]; methods-focused journals such as Statistics in Medicine [25]; textbooks that provide guidance with a more comprehensive scope. In addition,

What is new?

- Methods guidance (MG) supports the planning, conduct, and analysis of health research.
- Previous studies found a need for clarifying the possible approaches for developing MG.
- STrategies for developing REseArch Methods (STREAM) will identify such approaches and clarify when they are useful or not.
- The aims are to make guidance more transparent, evidence-based, and user-oriented.
- Developing STREAM involves reviews, interviews, panel discussions, and user testing.

policymakers, health authorities and regulators frequently publish MG. Well-known is the guidance developed under the auspices of the International Council for Harmonization [26] and adopted by the U.S. Federal Drug Administration [27], and the European Medicines Agency [28]. Furthermore, many scientific journals provide MG. For instance, in a sample of 165 health science journals, 122 (74%) provided statistical guidance as part of their instructions for authors [29].

Health researchers rely on these published MGs to inform optimal study design, conduct, analysis, and interpretation of health research studies. Funding agencies, policymakers, health authorities, regulators, journal editors, and peer reviewers rely on MG to defineexpectations against which they evaluate the quality of study protocols and reports [30–32]. Indeed, a survey among methodologists identified MG as a preferred way for disseminating and implementing research methods [21]. The fact that some MG articles are among the most frequently cited articles provides further evidence for their popularity [33–36].

Despite the importance of MG as a tool for promoting the use of appropriate research methods, best practices for developing MG have received little attention and remain largely unclear. Consequently, developers of MG use ad hoc methods or modify development processes originally designed for other purposes [37]. A number of studies have systematically reviewed MG articles and found that the majority of MG does not report a specific development process [37] and, if they do, they vary considerably in approaches to development [38]. Moreover, they rarely base the guidance on methodological evidence or input from guidance users, such as health researchers and other interest holders [39]. This lack of clarity on how to develop MG may affect their trustworthiness and practical utility [38–42] and may not lead to the optimal, patient-important evidence that is necessary to inform health care. For instance, a study that reviewed 13 guidance articles for psychiatric randomized clinical trials published by the Federal Drug Administration and European Medicines Agency identified serious deficits in recommendation for trial designs and transparency issues [40]. Another review of 57 guidelines for multivariable regression analysis found that most lacked advice on essential topics such as overfitting, interactions, missing values, or nonlinear predictors [41].

The aim of the present initiative is to describe STrategies for developing REseArch Methods guidance (STREAM) [43]. The strategies will support developers of MG in developing researcher-friendly and trustworthy MG to ultimately foster valid and relevant health research evidence.

2. Methods

2.1. Conceptual considerations

The specific focus on MG is relatively new. To provide a clear and transparent starting point for the development of STREAM, we propose a number of definitions and conceptual considerations that we may refine over the course of the project.

2.2. Definition of "methods"

For the purpose of STREAM, we define "methods" as the structured approaches and procedures, both statistical and nonstatistical (eg, methods for patient recruitment, causal diagrams, qualitative data analysis), that researchers employ in a study to collect, analyze, and interpret data to answer a research question.

2.3. Definition of "methods guidance"

For the purpose of STREAM, we define "methods guidance" (MG) as written information or practical advice aimed at (1) clarifying the concept of methods and underlying research challenges (eg, what is selection bias and which methods are available to prevent or address it?); (2) choosing between alternative methods (eg, what is the best method to handle missing data in a statistical analysis?); (3) applying a specific method (eg, how to perform a qualitative content analysis step-by-step); or (4) any combination of those three aims.

The definition does not encompass reporting guidelines and quality assessment tools, two other types of written guidance for health researchers. Although both can provide guidance on methods - reporting guidelines through defining the minimum content of methods sections, and quality assessment tools through highlighting important potential risks of biases - they do not provide explicit MG and, therefore, are not in the focus of STREAM. Table 1 summarizes key distinctions between these three types of guidance.

Table 1. Disambiguation between methods guidance, reporting guidelines, and quality assessment tools

Guidance type	Aim	Development process and format	Examples
Methods guidance [44]	To help researchers understand, select, and apply methods.	The development process is currently highly variable and mostly not reported; the presentation is highly variable [37].	[36,45–48]
Reporting guidelines [49]	To help researchers present methods, results, and other important aspects of a study in a manuscript.	A knowledge synthesis (eg, scoping review) typically followed by a Delphi study with a large group of interest-holders, resulting in a checklist with elaboration on each item [50].	[51–53]
Quality assessment tools [54]	To help researchers assess study methods and the credibility of results.	Variable development process, typically resulting in multidimensional measurement tools with items and a composite score, often with a manual [55].	[56–58]

3. The role of STREAM in the research process and key interest-holders

Considering the pathway from methods research to health care, STREAM will apply very early in that pathway (Fig 1). The target users of STREAM will be developers of MG. The developers will often collaborate with a panel that may involve methods experts, possibly early career methodologists (especially in the academic settings), users of MG such as health researchers, government bodies, journal editors, and funders among others (see Appendix A1 for a more detailed account of interest-holders in MG). Because the composition of teams developing MG can greatly differ, to ensure usability and uptake of STREAM, it should reflect the needs of methods experts but also first-time MG developers and developers with little methods training.

Health researchers in various roles, including clinical investigators, collaborating methodologists, editors, and reviewers, among others will not use STREAM but rather the MG developed using STREAM. To ensure relevance and uptake of such MG, STREAM should take into account the health researchers' needs and preferences too.

Ultimately, through optimizing MG and resulting clinical evidence, STREAM will have an impact on health care professionals and patients. While their experiences and preferences in specific disease areas can be crucial to inform health research (eg, research questions and outcomes), their involvement in methods research (such as this project) is more controversial and the subject of ongoing research [59]. To establish, if possible, a deliberative and meaningful contribution from patient partners, we will include an experienced patient partner on the steering committee.

3.1. Scope of STREAM

STREAM will address all stages of the development of MG ranging from assessing the need for MG to guidance publication and dissemination. Regarding study types, STREAM will address MG for any type of health research, including observational studies, clinical trials, qualitative studies, systematic reviews, or meta-analyses.

3.2. Usage and flexibility of STREAM

STREAM should serve as flexible guidance. It should be adaptable to MG for different topics, target audiences, and purposes. For instance, simulation studies may provide important evidence for quantitative but not qualitative MG. We acknowledge that not all steps and recommendations may always apply to all types of guidance, and that MG developed without any formal approaches, such as guidance based only on expert opinion, can still be very useful. This flexibility is also important to ensure that STREAM will be useful for the many MG developers that lack dedicated funding [37].

3.3. Format of STREAM

STREAM will be written guidance. It will have a clear structure and include actionable recommendations with rationales, real-world examples, and tools and templates to support practical application.

3.4. Dissemination

We plan to publish STREAM in open-access journal articles and maintain it as a living, that is, versioned open

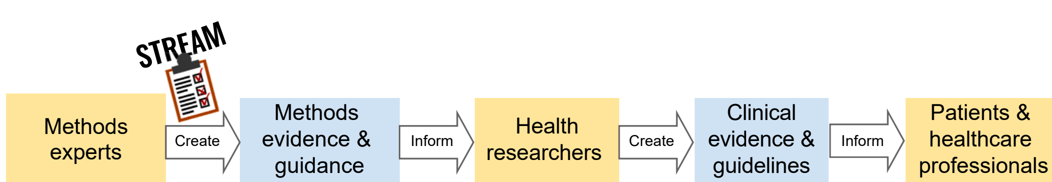

Figure 1. The role of STREAM in the process of informing methods research and indirectly health research and health care.

resource at https://lights.science/ to enable easy updates and provide supplementary tools and templates.

3.5. Compatibility with existing approaches and current practice of guidance development

To maximize acceptability, we will consider STREAM's compatibility with existing approaches to developing MG (to the extent established), such as the process from the STRengthening Analytical Thinking for Observational Studies (STRATOS) initiative [60], The Grading of Recommendations Assessment, Development and Evaluation (GRADE) working group [61], and European Medicines Agency [30,50] (we will search for any further existing approaches in work package 1). In addition, we will take into account the current practice of MG development (work package 2). Also, when developing STREAM, we will consider guideline development practices established in related research fields, such as reporting guidelines [50] and clinical practice guidelines [62].

4. Development process of STREAM

We plan for five work packages (Fig 2, discussed in detail below), which will involve different teams.

5. Teams

STREAM will involve a core group, a steering committee, and a consensus panel. In addition, work package 3 will involve a sample of users of MG, and work package 5 will involve a sample of developers of MG.

The core group includes M.C., S.S., M.B., and J.H. They will lead and carry out the individual work packages.

The steering committee includes 5 experienced developers of MG focusing on different areas, including clinical research and statistics in academia and the health care industry, qualitative research, reporting guidelines, quality assessment tools, and clinical practice guidelines (A.B., F.S., G.G., D.M., and F.B.) and 1 patient partner experienced in contribution to trial methods research (D.S.). The committee will meet regularly (3−5 times a year or, if necessary, more frequently) with the core group to refine the research plan, review progress, and resolve possible challenges. The steering committee agreed on this protocol and key conceptual considerations regarding the goal and scope of STREAM, acknowledging that the plan and concepts may evolve.

A consensus panel will consist of the steering committee plus 10-15 additional members including users of MG and other interest-holders (see work package 4). The panel will discuss the findings of work packages 1, 2, 3, and 5, may request additional research, develop an initial draft or STREAM, and conduct a consensus study to develop the final version of STREAM.

5.1. Work package 1: Scoping review of existing standards and suggestions for developing MG

Objective: To identify any existing standards and suggestions for developing MG for health research.

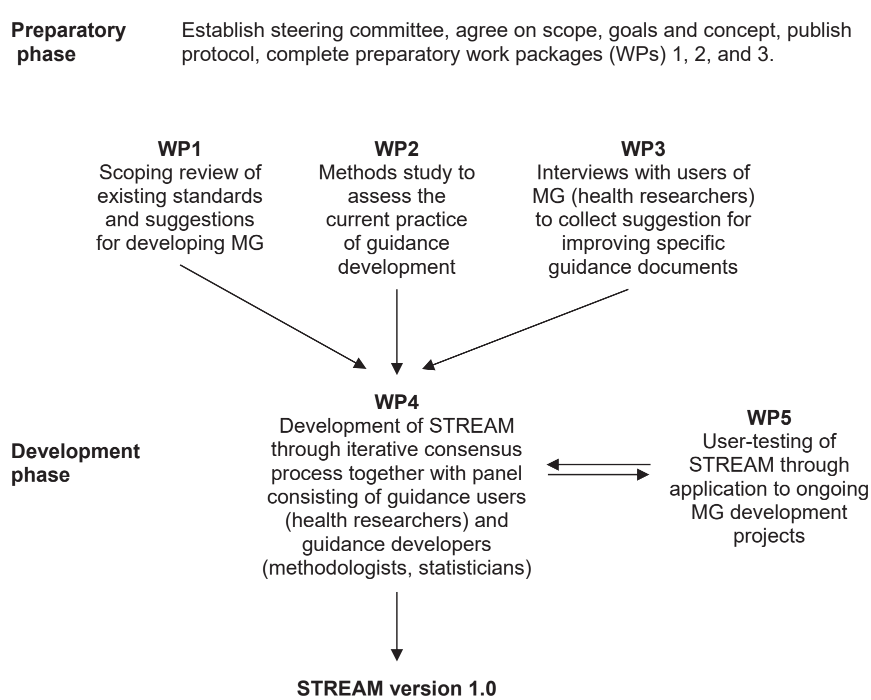

Figure 2. Overview of the development process.

Design: Scoping review [63−67] and qualitative content analysis [68].

Eligibility criteria: We will include publications that provide explicit standards or suggestions for developing MG for health research. The scope and focus of such eligible publications can differ and may include.

1. Development of MG, for example, guidance for developing articles from GRADE working group [61] and guidance for developing European Medicines Agency guidelines [30];
2. A study about MG, for example, a review of MG in a specific area.
3. Related topic, for example, guidance on methods review, meta-research, simulation study, and other possible inputs to MG.

We will accept any alternative terminology for MG including guideline, recommendation, how to, best practice, framework, and tutorial.

We will exclude publications clearly outside of health research as indicated by the scope of the journal, scope of the document, or type of examples discussed as well as documents that apply to reporting guidelines or quality assessment tools only.

Search and selection process: We will perform citation searching, beginning with 7 potentially eligible publications [38,50,55,61,69−71] that we identified through the screening process to inform the Library of Guidance for Health Scientists (LIGHTS) [72]. LIGHTS is an open-access database that we curate, for any type of MG in the context of health research. Based on the identified relevant publications, we will perform backward and forward citation searching to identify all cited and citing references using citationchaser [73,74]. Two reviewers will independently screen titles and abstracts for potential eligibility using Rayyan [75]. To assess definitive eligibility, one reviewer will use MAXQDA [76] qualitative analysis software and screen full texts and highlight any eligible sections. A second reviewer will double-check highlighted parts and screen for additional relevant sections. In addition to the initial citation search, we will search Google Scholar and use artificial intelligence-supported search tools (see Appendix A2). For any additional eligible publication identified through any of these search strategies, we will repeat backward and forward citation searching until no further eligible publications emerge. We do not plan to perform boolean searches in biomedical databases because of the low sensitivity and precision of such searches for methods topics [77].

Data extraction and qualitative analysis: We will apply manifest content analysis [68,78] using MAXQDA. We will focus on the sections that include suggestions and standards for developing MG. Two reviewers (one coding, the other checking) will code the relevant sections. We will start with an initial codebook that we developed based on our extensive experience in indexing of MG papers for the LIGHTS database (Appendix A3). We will modify, refine, and expand the codebook as needed [78].

We will use a common codebook for work package 1 and work package 2, as we expect that many codes will overlap. The common codebook may help us identify gaps in guidance or practice. For instance, we may identify specific methods for guidance development in work package 1 that none of the MG articles included in work package 2 applied in practice - or vice versa. The reviewers involved in work package 1 and work package 2 will meet regularly to discuss modifications of the codebook.

Another method to identify potential knowledge gaps is to chart the findings of work package 1 (and also work package 2 to identify gaps in practice) against formal guidance development processes that have evolved in other research areas such as the development processes of clinical practice guidelines [62] and reporting guidelines [50]. We may identify more and more suitable such processes when we review the publications in work package 1 and 2 and through discussions with the steering committee.

Purpose for STREAM: In work package 1, we will collect candidate elements and processes and identify needs and suggestions for the development of MG that other researchers previously expressed.

5.2. Work package 2: Methods study to assess the current practice of guidance development

Objective: To identify and better understand the spectrum of approaches that developers of MG use.

Design: Methods study of MG articles [79−83] and qualitative content analysis [68,78].

Sampling: We will draw a sample of MG articles from LIGHTS. Details on the search process are available at https://lights.science/. LIGHTS include MG that satisfies the following criteria.

1. A journal article or a guidance document from a regulatory authority.
2. Explicitly mentioned the aim to provide MG (accepting any alternative expression) in the title or abstract.
3. The MG addresses a topic relevant to health research in humans.

Specifically for this work package, we will add the following eligibility criteria.

4. The article includes a description of the development process of the MG;
5. Published between January 2020 to February 2025 (the last 5 years and approximately 100 articles; we may expand the time span if we need more articles.)

We will use a purposeful sampling approach to ensure a diverse sample of MG articles. Specifically, we will aim to stratify the sample according to main categories of methodological approaches (as outlined in Appendix A3), such as reviewing the methodological literature, performing a simulation study, involving interest-holders, or performing a consensus study. We aim to include at least 10 MG articles per broad category. As we progress with the coding (see next section), we may also add more articles to potential additional categories that we may identify.

Qualitative analysis: We will apply qualitative manifest content analysis [68,78]. The analysis will start using the same preliminary framework presented in Appendix A3. This framework will provide a preliminary structure for the codebook of the content analysis. We will first immerse ourselves in the data by thoroughly reading the methods sections of the included papers and conducting an initial round of coding on a broad level. Two researchers will split the sample of guidance papers and code the methods, processes, and people involved (and possibly other elements) in the development of the MG. They will use MAXQDA software [76] and the same shared codebook. They will then cross-check each other's coding and discuss codes and disagreements. This process will be iterative, allowing to going forth and back and refinement of the codes as new insights emerge. We will also chart the codes against other guidance development processes as mentioned in work package 1.

Purpose for STREAM: In the work package, we will identify candidate concepts, methods, and processes for STREAM. The work package will help us calibrate STREAM with current practice to support acceptance and feasibility. In addition, we will identify examples to support the discussions in work package 4.

5.3. Work package 3: Interviews with users of MG

Objective: To engage with users of MG to identify factors that make MG useful, trustworthy, and feasible as well as potential other facilitators and barriers to MG uptake.

Design: Semistructured interviews and qualitative thematic analysis [84,85].

Participants: We will include users of MG from 4 interest-holder groups (Appendix A1).

- Individual researchers from academia and industry who recently used MG ($n = 25$)
- Researchers working for funders or commissioners ($n = 5$)
- Researchers working for policymakers, regulators, and health authorities ($n = 5$)
- Journal editors ($n = 5$)

Sampling, sample size, recruitment: We will create a purposeful sample of MG users aiming to maximize variation. Apart from 4 interest-holder groups, we will additionally aim to balance the sample by the following factors: stratification factors: sex (equal number of men and women) [86], research experience (early career and experienced), sector (academia or industry, applies only to individual researchers), methods area (quantitative, qualitative, study management, implementation science), and geographical location. We will exclude participants of the core group, steering committee, or the panel. The users may include individuals who were previously involved in MG development (we will not check for it). To achieve a broad spectrum of researchers, we initially plan for around 40 interviews. This number is likely sufficient to reach data saturation regarding main facilitators and barriers (typical numbers range between 5 and 24, with a tendency for larger numbers in heterogeneous populations such as ours) [87]. If we fail to reach saturation as perceived by the interviewers (ie, no new important topics emerging in latest interviews), we will add more candidates.

We plan to recruit mostly individual researchers of which we are confident that they are frequent users of MG. We are reluctant to focus the interviews on employees of funders, government bodies, or publishers because we are uncertain whether they often consider themselves users of MG or not. If, during the interview study, we realize that they do, we may change the proportions.

When recruiting the 25 candidates from the first interest-holder group (individual researchers), we will link the candidates to specific MG that we reviewed in work package 2. For instance, we might use recent MG on topics such as literature search [88], statistical analysis planning [89], or on patient-reported outcome measures [90]. We will aim for diversity in terms of guidance topic, study type, difficulty of guidance, development process of the guidance, format, and journal. To achieve appropriate variation, we will suggest 50 guidance articles (or more if needed) to the steering committee and seek approval.

For each MG article, we will search the citing articles via citationchaser [74], order by publication date, and pick the most recent study, and review the study to check if the citation refers to a real application of the MG (rather than just citing it, for instance, in a discussion). We will retrieve the email address of the first author, statistician or methodologist (if any), and the last author. We will approach them as a group, ask for the person that has used and developed most experience with the cited MG, and invite that person to the interview. If the group suggests more than one individual, we will offer to interview them jointly.

Interview process: We will ask the participants to plan time shortly before the interview to refresh their memory regarding the specific MG document. The interview may take approx. 45 minutes. We will record the interview, with the permission of each individual, using video conferencing software and store them on encrypted and password-protected servers. Once transcribed, we will pseudonymize the transcript and delete the recordings.

Interview guides: We will prepare interview guides that include open-ended questions with the option for spontaneous in-depth exploration during the conversation (ie, semistructured) [91]. Appendix A4 shows an initial draft. We will tailor different versions to the individual interest-holder groups. Each interview will include questions about the participant (training, role, and expertise), about the strengths and limitations of the guidance document in general, and detailed questions addressing specifics of the MG linked to the candidate. The questions will explore facilitators and barriers to the use of MG informed by the Consolidated Framework in Implementation Research (CFIR) [92]. The relevant CFIR domain (domain one) provides 8 items [92]. Of those, 7 are relevant in the context of MG and will inform the interview guide (Appendix A4).

Analysis: The qualitative data analysis will start after the first interview, simultaneously with the data collection. We will use thematic analysis [84,85].

Purpose: To make sure that MG developed according to STREAM will align with the needs of guidance users to optimally support uptake and impact of MG.

5.4. Work package 4: Development of STREAM

Objective: To develop the new STREAM. Specific objectives include selecting the content for STREAM, finding a suitable structure, clarifying concepts and terminology, especially regarding methods evidence, develop approaches to presenting MG, define quality criteria for MG, develop templates and checklists for guidance developers, and develop approaches to MG implementation.

Design: Consensus study using the nominal group technique [93,94].

Consensus panel: The core group and the steering committee will work with a diverse panel representing relevant interest-holders (as specified in Appendix A1). They will include.

1. Individual researchers, research collaborations, and institutions ($n = 5$ in addition to the 4 members of the core group and 5 members of the steering committee that fall in this category): Members of the core group and the steering committee already represent MG developers from various universities and guidance development groups including the STRATOS initiative [60], the GRADE working group [45], the EQUATOR network [49], JBI [95,96], and Cochrane [97]. When selecting additional members for the panel, we will therefore focus on individuals working for the industry and users of MG.
2. Policymakers, regulators, and health authorities ($n = 3$): We will approach these organizations (see Appendix A1) for possible candidates with experience in commissioning, reviewing, developing, or applying MG.
3. Journal editors ($n = 3$): We will approach editors from journals that frequently publish MG, such as the British Medical Journal, Annals of Internal Medicine, Journal of Clinical Epidemiology, or Statistics in Medicine for possible candidates with experience in editing and reviewing but also developing MG as part of the journal's instructions for authors.
4. Funders and Commissioners ($n = 2$): We will approach these organizations for possible candidates with experience in commissioning, reviewing, developing, or applying MG.
5. Patient and public partners ($n = 1$): The steering committee already includes one patient partner. We will discuss together with the steering committee whether it makes sense to include additional partners.

To maximize variation, apart from the interest-holder group, we will consider additional factors as specified in work package 3: sex, research experience, sector, methods area, and geographical location. The sampling will be purposeful and also convenient as we may include individuals from our established professional networks.

In total, we plan to involve 23 members in the panel, a number which is within the typical range of participants for a discussion-based consensus study (the median number of participants across a sample of consensus studies using the nominal group technique was 16) [98]. We plan to involve mostly individual researchers as we anticipate challenges in recruiting representatives from other interest-holders. We may revise the proportion as we learn more about the interest holders during our work on the first three work packages.

Consensus methods: The nominal group technique provides methods for structuring group discussions, balancing input, determining how well participants agree on specific issues, and building group consensus [93,94]. We plan to use the following methods: discussion facilitated by a member of the core group who ensures that every participant gets the opportunity to speak; preparatory information package; record ideas privately and share them with the facilitator before meeting; online group meetings using video conferencing software. We anticipate that we may need 3-4 live discussions to develop a draft that is ready for user testing and another 2 meetings to discuss the feedback from the user testing and to finalize the recommendations. If necessary, we will arrange parallel meetings with interest-holders who can attend a meeting [99]. We will aim for consensus (defined as an agreement threshold of 80% or more of panellists, with the ones in disagreement feeling that the decision does not represent a serious mistake). We will adhere to reporting guidance for consensus studies [100].

Development process of STREAM: As a preparation for the first session, we will provide an information package that will include: a list with names and roles of the panellists, this protocol, a summary of the results of work packages 1, 2, and 3, and an early draft of STREAM. In the first live session, we will agree on the aims of the consensus study and the conceptual considerations; seek feedback on the results of work packages 1, 2, and 3 (with the possibility to request additional analyses); ask for interpretation of the results; and collect ideas for the structure and format of STREAM. The core group will document decisions, uncertainties, and further discussion points raised, built on the initial draft of STREAM, and share both the documentation and revised draft with the panellists and invite written feedback (and possibly send around again). In the following 2-3 sessions, the panel will discuss key issues as they emerge and achieve consensus to develop an advanced draft for user testing (work package 5 below). When the user testing is finished, the panel will convene again to decide about changes based on the user testing. Depending on the user-feedback, the panel may convene more than once during the user testing. The core group will create a final version and approach the panellists for final consensus. We may need additional rounds of written feedback and live discussions.

Purpose: To integrate the findings from work packages 1, 2, and 3, the preliminary conceptual considerations (see above), knowledge of panel members, generate new ideas, create a draft for testing in work package 5, and agree on a final version 1.0 of STREAM.

5.5. *Work package 5: User testing of STREAM*

Objective: To user-test STREAM in ongoing guidance development projects.

Design: Semistructured interview and thematic analysis [85].

Participants: We will approach leaders of ongoing MG development projects (at any stage, ranging from protocol development to finalizing) through the collaborators' professional networks. In addition, we will approach the following groups that regularly develop MG: European Medicines Agency; Federal Drug Administration; Patient-centered outcomes research institute; STRATOS initiative; Trial forge; the COMET initiative; GRADE; Cochrane; JBI; Campbell Collaboration. We may identify additional groups and individuals in work package 1 and work package 2. Further, we may find published protocols for MG projects through the searches for LIGHTS. Should we identify more MG development projects than needed, we will invite groups in an order to introduce maximum variation regarding guidance characteristics (statistical/nonstatistical, academic/regulatory, guidance type).

Sample size: We aim to interview at least 12 individuals (potentially sufficient to reach saturation) [87], ideally from 12 different ongoing guidance development projects. We will first invite the leaders of the guidance development projects and, if not available, ask for a replacement. If more than one group member is interested, we may arrange a focus group instead of individual interviews.

Process: We will share a draft of STREAM with the candidate along with the instruction to share STREAM with their team members, discuss its implementation, and apply it. We will arrange a date for the interview 4 weeks after this initial contact. We may send reminders of the ongoing user test after 2 weeks, including an offer to postpone the interview if the participants need more time to develop experience with STREAM. We will develop an interview guide following the principles outlined in work package 3. The guide will include questions addressing the overall experience with STREAM and then focus on the elements of STREAM that the participant or the group has developed most experience with. Then, we will seek feedback on the remaining elements of STREAM. After each interview, we will offer to support the groups in applying STREAM, possibly resulting in additional experiences through collaboration. We will record, transcribe, and analyse the interviews (ca. 45 min) as explained in work package 3. We will record positive feedback, challenges, and suggestions for improvement. We plan for 1-2 additional meetings with the panel to develop the final version of STREAM based on the user testing.

Purpose: To collect feedback from STREAM users regarding their needs, perceived barriers and suggestions for improving STREAM, and to refine STREAM based on that feedback.

6. Discussion

The initiative to develop STREAM will address the current uncertainty regarding possible processes and approaches for developing MG. It includes several innovations. Conceptualizing MG as an integral part of the evidence production and implementation pipeline is novel. Through defining the common goal of developing MG and developing concepts of methods evidence, STREAM will link together subareas of methods research that are currently poorly connected such as meta-science, simulation studies, and comparisons of methods.

The guidance will be primarily for methodologists who are considering using formal methods for developing MG. STREAM will ensure that their guidance development process is based on sound principles, potentially increasing the credibility of their work. Clarifying the possible approaches may also help establish a common understanding of best practices in MG development in the research community. STREAM will facilitate structured development processes and harmonize concepts and terminology, including methods evidence, guidance users, and guidance types. In addition, STREAM may lead to more structured reporting of MG and could be complemented with formal reporting guidelines for MG. Further, we may establish quality criteria for MG, which could eventually lead to a formal quality assessment tool for MG. Finally, some of the knowledge that we will produce for STREAM might also inform the development processes of reporting guidelines and quality assessment tools.

There are potential concerns. Since most MG does not follow a formal development process, some may ask if the introduction of STREAM could imply that existing guidance developed without a methods section is invalid. We clarify that STREAM does not intend to suggest that past or future guidance developed through other or informal approaches is less valid. We clarify that this is not the case. Rather, STREAM will offer coherent considerations for researchers who want to develop MG in a systematic way. Also, we acknowledge that applying the new strategies may not always lead to high-quality MG. Quality will also strongly depend on the knowledge, experience, and communication abilities of the guidance authors. Although the development process of STREAM may identify some quality criteria for MG, especially if users suggested them, STREAM will not serve as a quality framework. Developing such criteria could be a possible future step.

We anticipate a number of challenges: one is the lack of a universally accepted development process for MG. Considering that STREAM itself will be MG, the principles developed through this process will also apply to it. Rather than considering this a strictly prespecified protocol, updating the protocol as STREAM evolves could benefit the project.

Another challenge is the involvement of interest-holders. While the interest-holders in health research are well-established (see Appendix A1), identifying interest-holders in MG and defining their specific roles in the work packages is not straightforward and requires learning. For instance, although patients will probably not use STREAM or resulting MG, they are sometimes involved in methods research [59]. To establish whether and how patient and public involvement and engagement can contribute to the development process of STREAM, we will work with an experienced patient partner (D.S.) throughout the work packages.

Another challenge is evaluating the uptake and effectiveness of STREAM in practice outside the 4-year timeframe to develop STREAM. Although the development will include the testing usefulness of STREAM, it will take time, possibly 5 to 10 years, before we can evaluate the uptake and impact of STREAM based on an analysis of resulting MG and qualitative feedback based on a larger number of STREAM users and the users of MG developed according to STREAM. One possibility to expedite such real-world feedback is to publish preliminary versions of STREAM already early in the project. In addition, to support uptake, we plan for implementation activities such as workshops with guidance developers and at methods conferences.

The rapid evolution of artificial intelligence may present additional challenges. The attitudes of researchers toward written MG may change. Instead of retrieving peer reviewed publications, researchers may increasingly rely on MG generated by large language models (LLMs). It could therefore be important to consider possibilities how STREAM could influence LLM's approach to guidance generation. In addition, a section of STREAM that addressed implementation of guidance may outline principles to optimize MG developed according to STREAM for uptake by LLMs. Standardized reporting, using clear terminology, and explicit communication of certainty may be important for such principles.

Although the development of STREAM will be conceptually challenging and possibly require substantial time, a number of factors support its feasibility. One is the fact that very little systematic research has happened in the space of MG thus far, leaving a large knowledge gap to fill. This will likely allow for a relatively great flexibility with respect to the scope and level of detail. In other words, when aspects of STREAM may prove controversial or in need of further research outside the scope of this initiative (eg, how to optimally present guidance), we could still make useful contributions through providing preliminary solutions or directions for future research. Another factor supporting the feasibility of STREAM is the consideration of examples from related research guidance. In the area of clinical practice guidelines, the approach from the GRADE working group, initially proposed around 25 years ago, now adopted by many guidelines groups, greatly improved the methodological rigor, transparency, and relevance of clinical practice guidelines [35]. Similarly, a development process for reporting guidelines from 2010 that proposed literature reviews, the involvement of interest-holders, Delphi studies, and principles for publication and dissemination has led to many reporting guidelines developed in a similar fashion and presented an established checklist format [50]. Another good example is the establishment of standards for core outcome sets and associated reporting guidelines and tools for quality evaluation [55,101]. All the examples show how a research community formed around shared methods and how having these approaches in place for long enough can lead to clear improvements in the resulting guidance.

CRediT authorship contribution statement

Malena Chiaborelli: Writing – review & editing, Writing – original draft, Project administration, Methodology. **Julian Hirt:** Writing – review & editing, Writing – original draft, Methodology. **Matthias Briel:** Writing – review & editing, Methodology. **Frances Shiely:** Writing – review & editing, Methodology, Conceptualization. **Anne-Laure Boulesteix:** Writing – review & editing, Methodology. **Gordon Guyatt:** Writing – review & editing, Methodology, Conceptualization. **Frank Bretz:** Writing – review & editing, Methodology. **David Moher:** Writing – review & editing, Methodology. **Derek Stewart:** Writing – review & editing. **Stefan Schandelmaier:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

Stefan Schandelmaier reports financial support was provided by Swiss National Science Foundation. Frank Bretz reports a relationship with Novartis that includes: employment, equity or stocks, and travel reimbursement. - Stefan

Schandelmaier is receiving funding from the Swiss National Science Foundation (Ambizione grant number 223702) with the expectation to develop parts of his career around the topic of methods guidance. - Frank Bretz reports a relationship with professional societies, such as the American Statistical Association, that includes travel reimbursement. - Gordon Guyatt is Chief methodologist and board member of the MAGIC Evidence Ecosystem Foundation (www.magicproject.org), a non-for profit organization, which conducts research and guideline methodology and implementation, and which provides a authoring and publication software (MAGICapp) for evidence summaries, guidelines and decision aids. GG has not derived any income from this work. - Given his role as a member of the journal's editorial board, David Moher had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2026.112184.

Data availability

No data was used for the research described in the article.

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